SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY (SLIIT)

ETHICS REVIEW COMMITTEE (ERC)

APPLICATION FOR ETHICAL APPROVAL

Please read the guidance notes carefully before completing this form.

This form must be completed electronically and sent to the Administrative Secretary of the ERC via email (**SLIITERC@sliit.lk)**. You must NOT start the fieldwork or data collection until you have been informed that your application has been given permission to proceed.

SECTION 1: DETAILS OF APPLICANT/S

Please provide the information about the Principal Investigator and all Co-investigators in this section. **Supervisor** should be added as the Principal Investigator in all student research.

Principal Investigator (PI):		
Name:	Prof. Samantha Thelijjagoda	
SLIIT Staff / Student ID:		
Faculty/School:	Faculty of Computing, SLIIT	
Role in the Department:	☐ Undergraduate Student ☐ Postgraduate student	☑ Staff member ☐ Postdoctoral Researcher ☐ Other (please specify)
Email:	samantha.t@sliit.lk	
Contact No.:	0774402550	
Co-Investigator 1 (if applicable):		
Name:	Dayana Priyadharshani Kumar	
SLIIT Staff / Student ID:	IT22178640	
Faculty/School:	Faculty of Computing, SLIIT	
Role in the Department:	☑ Undergraduate Student ☐ Postgraduate student	☐ Staff member ☐ Postdoctoral Researcher ☐ Other (please specify)
Email:	it22178640@my.sliit.lk	
Contact No.:	0764043970	
Co-Investigator 2 (if applicable):		
Name:	Harishalinee Elangovan	
SLIIT Staff / Student ID:	IT22057488	
Faculty/School:	Faculty of Computing, SLIIT	
Role in the Department:	☑ Undergraduate Student ☐ Postgraduate student	☐ Staff member ☐ Postdoctoral Researcher ☐ Other (please specify)
Email:	it22057488@my.sliit.lk	
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Co-Investigator 3 (if applicable):		
Name:	Vidursha Prabagaran	
SLIIT Staff / Student ID:	IT22294098	
Faculty/School:	Faculty of Computing, SLIIT	
Role in the Department:	☑ Undergraduate Student ☐ Postgraduate student	☐ Staff member ☐ Postdoctoral Researcher ☐ Other (please specify)
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Co-Investigator 4 (if applicable):		
Name:	Kaushalya Nagenthraraja	
SLIIT Staff / Student ID:	IT22289384	
Faculty/School:	Faculty of Computing, SLIIT	
Role in the Department:	☑ Undergraduate Student ☐ Postgraduate student	☐ Staff member ☐ Postdoctoral Researcher ☐ Other (please specify)
Email:	it22289384@my.sliit.lk	
Contact No.:	0764232082	

SECTION 2: GENERAL INFORMATION ABOUT THE PROJECT

2) Title of the Project:

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery (Project ID: R26-DS-010)

3) Dates and locations of proposed field work:

August 2026 (various weeks throughout the month) in-house webcam fall-scenario recordings were conducted at a private residence (home setting). Caregiving ADL activity observation and Excel logging were conducted at Love and Care Elders' Home, Colombo 13, Sri Lanka.

4) Please provide a lay summary of your project. Include background, aims/objectives, methodology, data analysis methods and expected outcomes in plain or non-technical language

Caregivers looking after elderly or unwell family members and patients often face heavy workloads, making it hard to keep track of daily care activities, monitor safety risks such as falls, manage their own stress, and communicate effectively (including with sign language). This project, CareSense, builds an integrated AI-based support system with four parts:
(1) Predicting caregiver stress from wearable and behavioral data
(2) Detecting fall risk in real time from ordinary webcam video
(3) Automatically turning a caregiver's spoken notes into structured daily activity records
(4) Teaching sign language to caregivers through an interactive tutor.
Most data used comes from existing public research datasets. Two parts of the project additionally use data collected directly by the team: short videos of consenting volunteers simulating falls, and a log of daily caregiving activities recorded at a caregiver's home. Data is analyzed using machine learning models, and results are reported as accuracy and safety-relevant metrics. The expected outcome is a working prototype demonstrating how AI can reduce caregiver burden and improve patient safety.

5) Describe the impact that your research will have on the community

The system aims to reduce documentation burden and stress on caregivers, improve early detection of fall risk for vulnerable patients, and support communication training with potential benefit to elder-care facilities, home caregivers, and care recipients in Sri Lanka.

SECTION 3: PARTICIPANT RECRUITMENT AND CONSENT

6.) Are you recruiting participants? (i.e. collecting data directly from participants through interviews or survey ?)	☑Yes ☐No (proceed to Q 14)
7.) Do your participants include SLIIT students or staff?	☐Yes ☑No

8.) Describe your target population and sample size

3 participants for live end-to-end testing of the fall-detection system via webcam recording: the researcher (Harishalinee Elangovan (Group Member), a SLIIT student, who organized and conducted the recording) and two family members (mother and brother of Harishalinee). These recordings were used only for qualitative live/system testing, not as training data — model training relied entirely on public benchmark datasets (NTU RGB+D, UR Fall Detection). Separately, for the ADL-extraction component, caregiving activity data was observed and logged at Love and Care Elders' Home, Colombo 13 (an elder-care institution), covering its resident/caregiving population; this data collection visit was conducted jointly by two group members, Vidursha Prabagaran and Harishalinee Elangovan.

9). Describe your recruitment method and process:

Webcam fall-detection testing: direct personal recruitment (self and family members), organized solely by Harishalinee Elangovan, for live system testing, not for training data. ADL-extraction data: obtained through institutional cooperation with Love and Care Elders' Home, Colombo 13, with permission from supervisor this data collection was carried out by Vidursha Prabagaran.

10.) Will participants be paid or receive anything in exchange for participating in the research?	☑No ☐ Yes
11.) Will you obtain consent from participants?	☐ No ☑Yes

If "No" (Explain why consent is not obtained and how privacy and confidentiality will be maintained)

N/A

If "Yes" (Describe how consent will be obtained and recorded. E.g. hard copy, email, online form, etc.)

For the webcam fall-detection test recordings: informed consent was obtained directly from all three participants (the researcher and two family members) prior to filming, as they are known personally to the researcher. For the ADL activity logging at Love and Care Elders' Home, Colombo 13: permission was obtained from the facility at the time of data collection; the team is preparing a formal written permission/consent record from facility management (and information sheet covering residents' data) to document this properly before the data is used further or published.

12.) Is there a potential for the participants to feel pressured to give consent to participate in the research due to position of power in dependent/unequal relationships?	☑No ☐ Yes

If "Yes" (Describe the relationship between the researcher and the participants, and how this issue will be addressed): N/A

13) Will you collect identified information (e.g: health information) **either from the government or private sector institutions without consent?**	☑No ☐ Yes

If “Yes” (provide details of what information will be obtained, from what organization, how privacy, confidentiality and anonymity will be secured): N/A

14) Does your research involve any of the following: ☐ NO – Proceed to Section 5 ☑YES – Proceed to Section 4	☐ Participants under 18 years of age ☐ Participants unable to give consent (e.g. cognitive impairment) ☐ Prisoners or people in custody ☑Elderly Persons ☐ Pregnant women ☐ People involved in illegal activities ☐ Deception of participants ☐ Sensitive topics ☐ Collection of genetic or bio samples

SECTION 4: VULNERABLE PARTICIPANTS, SENSITIVE TOPICS ETC

If you ticked YES to any of the categories in Question 14, you must provide details below. If you ticked NO, leave this section blank.

15.) Does your research involve participation in children and young people?	☑No ☐ Yes
16.) Does your research involve participation of other vulnerable groups such as pregnant women, those suffering from cognitive impairment, intellectual disability, mental illness, prisoners?	☑No ☐ Yes
17.) Does the research use deception, concealment, incomplete disclosure, opt-out approach or use of information without obtaining specified consent from those participants? Note: *Concealment/incomplete disclosure is defined as not fully disclosing or describing the aims or methods of observational research in public contexts. An opt-out approach is where information is provided to the potential participant regarding their participation is presumed unless they take action to decline to participate.*	☑No ☐ Yes
18.) Does your research involve people who may be involved in illegal activities?	☑No ☐ Yes
19 Does your study involve collecting data on sensitive topics? (E.g. topics such participants' sexual behavior, illegal or political behavior,	☑No ☐ Yes

experience of violence, abuse or exploitation, mental health, or gender or ethnic status etc)	
20 Does your study involve collecting genetic or biological samples?	☑No ☐ Yes

SECTION 5: RISK ASSESSMENT

21Risks to Participants

Outline the potential risks for participants. Explain how this research justifies the burden and risk to participants (*illness or injury, side effects, embarrassment, economic loss, exposure to prosecution, anything stressful, noxious or unpleasant, and complaints. Ensure to address in the Information Statement.*), and the steps taken to mitigate the risks (*e.g., Provide resources for counseling, allow withdrawal, use password encryption, anonymize data)*.

Risk Description	Justification	Mitigation Strategy
Re-identification of the caregiving household/care recipient from ADL activity logs	Realistic, real-world caregiving data is necessary to validate the ADL-extraction model beyond synthetic/public data	Data will be coded (participant/patient identifiers), anonymized before analysis and publication, and access restricted to the responsible researcher
Identifiability of volunteers in webcam fall-detection recordings	Realistic webcam footage is required to validate fall-detection performance beyond public lab datasets	Written informed consent obtained prior to recording; footage stored securely and not published in raw/identifiable form
Sensitivity of health/medication data collected for ADL logging	Medication, symptoms and mobility data are core to the ADL-extraction and alerting use case	Coding/anonymization, password-protected storage, and restricted access to raw data

22 Risks to Researcher(s)

Outline the potential harm or risk this research exposes to the research team (for eg.: dangers to personal safety), and if identified how these will be mitigated and your plan of action should these risks occur. (*e.g., Traveling to remote location for interviews)*

Hazard / Risk Description	Mitigation Strategy/Plan of action
Personal safety while visiting Love and Care Elders' Home to observe/collect ADL activity data	Visit conducted jointly by two group members (Vidursha Prabagaran and Harishalinee Elangovan) with prior arrangement/permission from facility management; visit limited to facility operating hours
No other significant risks to researchers identified (remaining data collection was performed on public datasets or standard webcam recording in a controlled setting)	N/A

23 Risks to the University and the Research

Outline any risks to the University (e.g. reputational damage if the topic is controversial) and the research (e.g: loss of data, inability to secure an adequate number of participants etc.) and the mitigation strategies

Hazard / Risk Description	Mitigation Strategy/Plan of action
Reputational risk to the University if sensitive health/medication data is mishandled or disclosed	Strict adherence to coding/anonymization, password protection, and restricted access as outlined in Section 7; ethics approval sought prior to further use
Risk of insufficient real-world data limiting research validity	Mitigated by combining the small in-house/primary datasets with much larger public benchmark datasets (TILES-2018, NTU RGB+D, UR Fall Detection, Svarah, WLASL, ASL Citizen)

SECTION 7: DATA MANAGEMENT & SECURITY

24 Data Collection Method:	☐ Online Survey ☐ Paper-based ☐ Interview ☑ System Logs ☑ Other:

25 Data Storage: Where will the data be stored?

Data will be stored on the personal laptops/devices of the four group members, protected by password and full-disk encryption. No third-party cloud storage is used for raw/identifiable data. A single anonymized/coded copy of processed data will be retained in an institutional repository if required for submission.

26 Data Retention and Destruction:

Data will be retained until project completion and final grading (approximately 6 months), after which it will be destroyed as described below.

27 How will the data be destroyed after this period?

Raw and identifiable data (webcam footage, uncoded ADL logs) will be permanently and securely deleted from all student devices. Only a single anonymised/coded copy will be retained in the institution's repository, in line with FoC data storage requirements

SECTION 8: CONFLICTS OF INTEREST

28 Are there any potential conflicts of interest?	☑ No ☐ Yes

SECTION 9: SUPPORTING DOCUMENTS CHECKLIST

☑Participant Information Sheet
☑Informed Consent Form
☐ Questionnaire / Interview Guide
☐ Translations of questionnaire etc. where different languages are used
☐ Recruitment Materials (advertisements etc. if applicable)
☑Any other relevant information/document

SECTION 10: DECLARATIONS & SIGNATURES

I certify that:

1. The information provided in this form is accurate to the best of my knowledge.
2. I will conduct the research strictly in accordance with this proposal and any conditions imposed by the ERC.
3. I will report immediately to the ERC any serious adverse events or ethical issues arising during the research.
4. I am familiar with the SLIIT Code of Ethics for Research.

	Name	**Signature**	**Date**
Principal Investigator	Prof. Samantha Thelijjagoda		
Co-Investigator 1	Dayana Priyadharshani Kumar		27.08.2026
Co-Investigator 2	Harishalinee Elangovan		27.08.2026
Co-Investigator 3	Vidursha Prabagaran		27.08.2026
Co-Investigator 4	Kaushalya Nagenthraraja		27.08.2026

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ERC Reference Number:	
Review Outcome:	☐ Approved ☐ Approved with Amendments ☐ Rejected
Comments:	
ERC Chairperson Signature & Date:	